PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM | CYBERSECURITY | NETWORKWALKS

Pentester Name	Oluwafunbi Bader
Program / Batch	B084 – Networkwalks
Report Date	11 November 2026
Modules Completed	W2-PM1 (Multiple Kali Tools), W2-PM5 (Zenmap Scanning)
Client / Target	networkwalks.com (written permission secured); my own local network (10.60.174.0/24)
Permission Secured from Client?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting; Phase 2: Scanning & Network Discovery; Phases 3–5: in progress

1. Liability Disclaimer

I performed all activities in this report only on systems and devices for which I held written permission, or which I own myself. These materials are for education and research purposes only and must not be used to break the law. The instructor, the authors and Networkwalks are not responsible for what I do with this knowledge. Every action I take is my own responsibility. Misuse can lead to criminal charges, heavy fines, loss of my job and a permanent record. In most countries, unauthorised access is a crime even when no damage is done.

2. Introduction

This report covers footprinting the networkwalks.com domain with multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

The footprinting commands were run in Kali Linux, running as a VirtualBox virtual machine on a Windows PC. The scanning was done with Zenmap on the Windows PC. Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux (VirtualBox) & Windows	Operating systems used for the reconnaissance and scanning activities
WHOIS	Find domain registration details (registrar, dates, name servers)

Tool	Purpose
whatweb	Fingerprint web technologies (server, CMS, plugins, IP)
nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers of the website
wafw00f	Detect whether a Web Application Firewall protects the site
dnsrecon	Enumerate DNS records (SOA, A, SRV, DNSSEC status)
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses, and draw the topology

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against networkwalks.com on 8 November 2026 using six Kali Linux tools. Each tool collected a different type of information about the target.

WHOIS (`whois networkwalks.com`). The domain is registered through GoDaddy.com, LLC (IANA ID 146). It was created on 6 November 2019, last updated on 12 November 2025 and expires on 6 November 2027. The name servers are NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM, which shows the site's DNS is handled by HostGator. DNSSEC is unsigned, and the domain carries the client delete, renew, transfer and update prohibited locks. The output also notes that this WHOIS server is being retired in favour of RDAP.

WhatWeb (`whatweb networkwalks.com`). The HTTPS site returned 200 OK and was identified as running Apache, WordPress 7.1.4 and the WordPress Download Manager plugin 3.3.76, with jQuery 3.7.1. It also exposed the server IP address (192.232.216.135), a contact email (info@networkwalks.com), an HttpOnly cookie named `__wpdm_client` and the page title "Networkwalks Academy". The plain HTTP request failed with "execution expired", so only the HTTPS result was returned. WhatWeb also listed Bootstrap with the same version number as WordPress, which looks like a detection quirk and would need manual verification.

Nslookup (`nslookup networkwalks.com`). The domain resolved to 192.232.216.135 as a non-authoritative answer. My Kali VM sent the query to its configured DNS server, 10.60.174.106, and the follow-up attempts to that server timed out, which is why the output ends with "no servers could be reached".

Curl (`curl -I https://networkwalks.com`). The response was HTTP/2 200 from an Apache server, dated Sun, 08 Nov 2026 11:26:07 GMT. The headers exposed the WordPress REST API endpoints (`/wp-json/` and `/wp-json/wp/v2/pages/53`), a `__wpdm_client` cookie set with the Secure and HttpOnly flags, and caching headers (`x-nginx-cache: WordPress`, `x-endurance-cache-level: 0`) that hint at the hosting stack.

Wafw00f (`wafw00f networkwalks.com`). Wafw00f v2.3.1 used 2 requests and reported that the site is behind a ModSecurity (SpiderLabs) Web Application Firewall.

DNSRecon (`dnsrecon -d networkwalks.com`). DNSRecon confirmed DNSSEC is not configured and returned the A record 192.232.216.135. It found no SRV records. The SOA lookup failed with "The DNS operation timed out" against 10.60.174.106, so name server, mail server and TXT/SPF records were not collected in this run.

4.2 Network Scanning with Zenmap

For the second activity I used Zenmap on 11 November 2026 to discover live hosts on my local network. I targeted the subnet 10.60.174.0/24, selected the Ping scan profile and ran `nmap -sn 10.60.174.0/24` (Nmap 7.993 as shown in Zenmap).

The scan checked 256 IP addresses and found 2 hosts up in 4.65 seconds:

- **10.60.174.106** is up with 0.013 s latency and MAC address A2:51:2B:1B:27:89, vendor unknown. My Kali VM uses this address as its DNS server, so it is most likely the gateway or router. The MAC address is a locally administered (randomised) type, which is common on phone hotspots and virtual devices.
- **10.60.174.211** is up, with no MAC address shown. Nmap does not print a MAC for the machine it runs on, so this is most likely the Windows PC running the scan.

The Hosts panel lists two more entries, 192.168.6.1 and host.docker.internal (10.2.0.2). They are not part of this ping scan's output and most likely come from other network interfaces on the same PC (for example a virtual or Docker adapter).

I then opened the Topology tab. It shows localhost at the centre, linked to 10.60.174.106, 10.60.174.211, 192.168.6.1 and host.docker.internal (10.2.0.2). I saved the topology as a PDF as the practical requires.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	**Web technology and version information exposed**	WhatWeb identified WordPress 7.1.4 and WordPress Download Manager 3.3.76	Attackers can use exposed software and version details to decide what to review for known weaknesses	**Medium**
2	**Contact email exposed**	WhatWeb returned info@networkwalks.com from the page	Can be used for phishing or social-engineering attempts against the organization	**Low**
3	**Server IP address identifiable**	Nslookup and WhatWeb resolved the domain to 192.232.216.135	Shows the network location of the web service	**Low**
4	**HTTP technical information exposed**	Curl returned headers revealing the WordPress REST API (/wp-json/), Apache and caching details	Helps technology fingerprinting and further enumeration	**Low**
5	**WAF technology identifiable**	Wafw00f identified ModSecurity (SpiderLabs)	Reveals part of the web application's security architecture	**Low**
6	**DNSSEC not configured**	WHOIS showed DNSSEC unsigned and DNSRecon confirmed it is not configured	DNS answers for the domain cannot be cryptographically verified, which leaves room for DNS spoofing attacks	**Low**
7	**Live hosts visible on the local network**	Zenmap found 2 live hosts (10.60.174.106 and 10.60.174.211) in 10.60.174.0/24	Any unexpected device on a network should be identified and verified	**Low**

Risk level key: Critical Medium Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. These practical exercises involved only information gathering and host discovery; no exploitation or vulnerability validation was performed. The presence of a software version, IP address or DNS record does

not by itself mean the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

Two limits apply to the results. The DNS server my Kali VM used (10.60.174.106) timed out during the nslookup and DNSRecon runs, so the name server, mail server and TXT/SPF records were not collected. The ping scan also identified hosts only; it did not scan ports or services.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements.

1. **Review publicly exposed technology information.** Regularly check what the website reveals about its CMS, plugins and versions, and hide version details where possible.
2. **Keep software updated.** Update WordPress, the Download Manager plugin and other components regularly, and review them against current security advisories.
3. **Review HTTP headers.** Check response headers and the exposed REST API (/wp-json/) to confirm that unnecessary technical information is not being revealed.
4. **Enable DNSSEC and review DNS records regularly.** Sign the domain with DNSSEC where the DNS host supports it, and check periodically that only required records are public.
5. **Properly configure and monitor the WAF.** Keep ModSecurity enabled and tuned, since it already blocks naive attacks.
6. **Limit publicly listed contact details.** Use a monitored shared mailbox, or a contact form, instead of publishing a direct address in page content.
7. **Perform regular internal network discovery.** Scan your own networks periodically to identify active devices.
8. **Investigate unknown devices.** Any unexpected device found during scanning should be identified and verified.
9. **Maintain network documentation.** Keep the network topology and device information documented and up to date.
10. **Perform security testing with authorization.** Run reconnaissance and scanning only against systems and networks where proper authorization has been given.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about networkwalks.com. WHOIS gave registration and name server details, WhatWeb identified the web technologies, Nslookup resolved the domain to its IP address, Curl exposed the HTTP headers, Wafw00f identified the WAF and DNSRecon showed that DNSSEC is not configured. In the network scanning activity, I used Zenmap to run a ping scan of my local subnet, discovered the active hosts and generated a network topology.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a lot about an environment from public information and network responses. I also learned that tool output has limits: when my DNS server timed out, some DNSRecon results were missing, and I had to document that honestly instead of assuming what the records contained.

Finally, I learned that a good report explains what was performed, what was found, what it means, what risk it may create and how to reduce that risk, and that reconnaissance and scanning must always stay within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

Figure 1: WHOIS lookup of networkwalks.com (`whois networkwalks.com`, 8 November 2026)

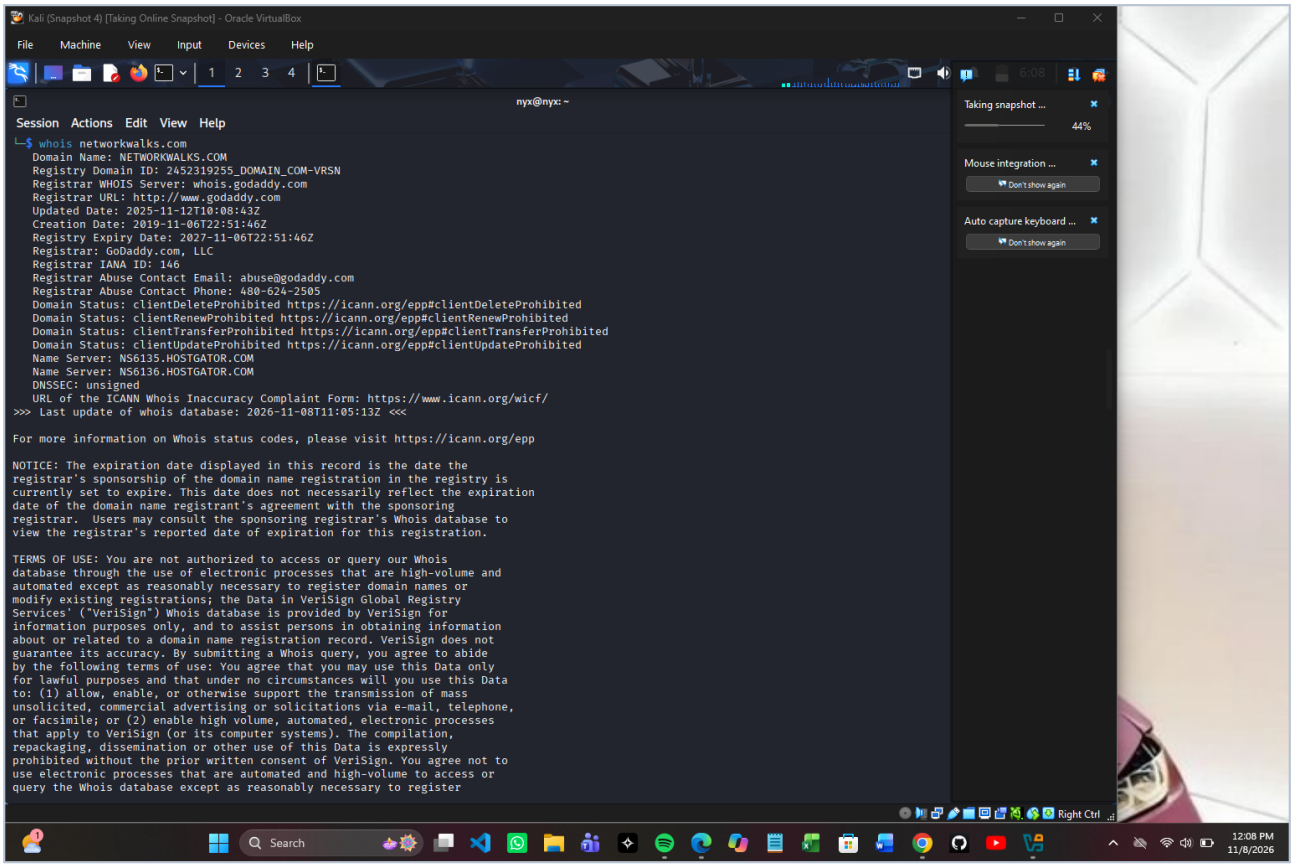


Figure 2: WhatWeb and Nslookup results (`whatweb networkwalks.com`, `nslookup networkwalks.com`)

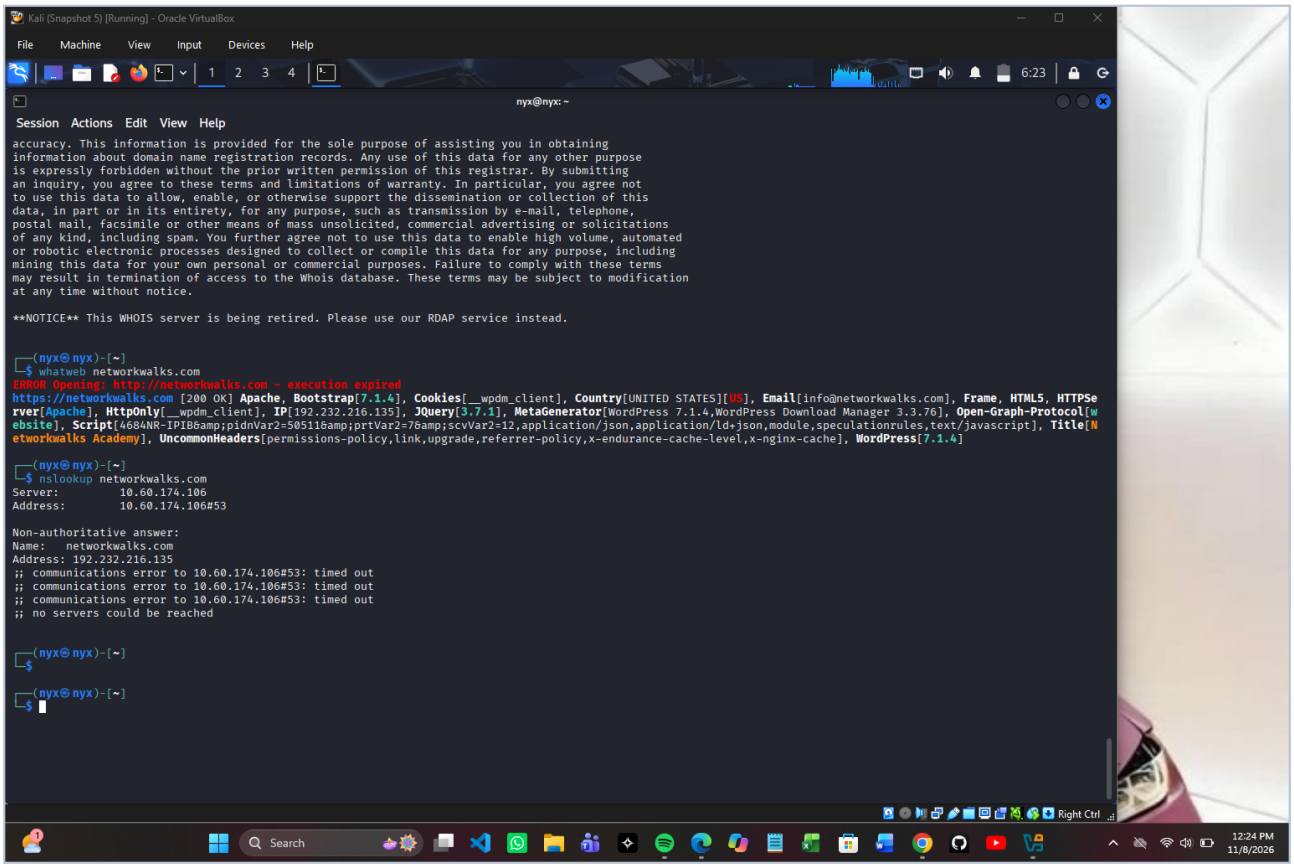


Figure 3: Curl response headers and Wafw00f WAF detection (`curl -I https://networkwalks.com`, `wafw00f networkwalks.com`)

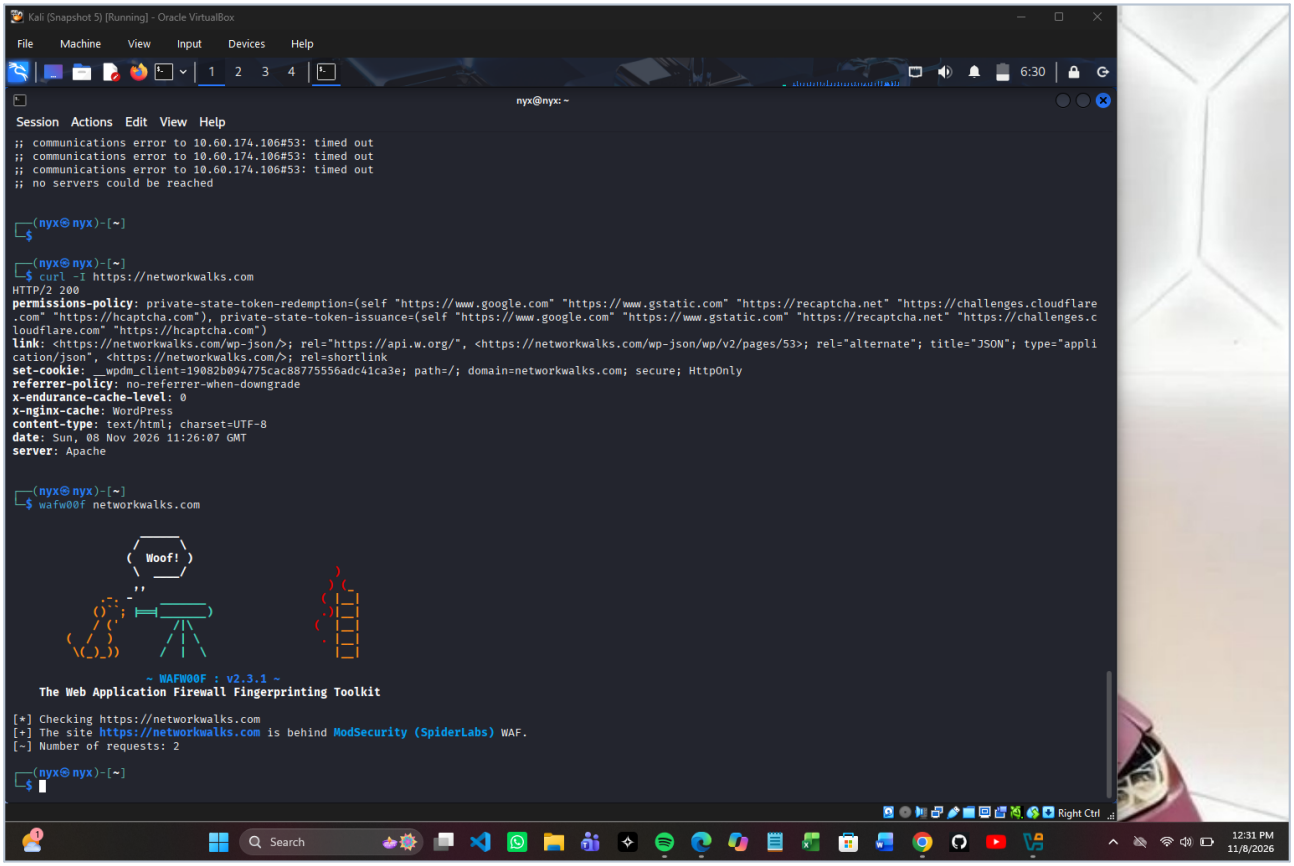


Figure 4: DNSRecon enumeration (`dnsrecon -d networkwalks.com`)

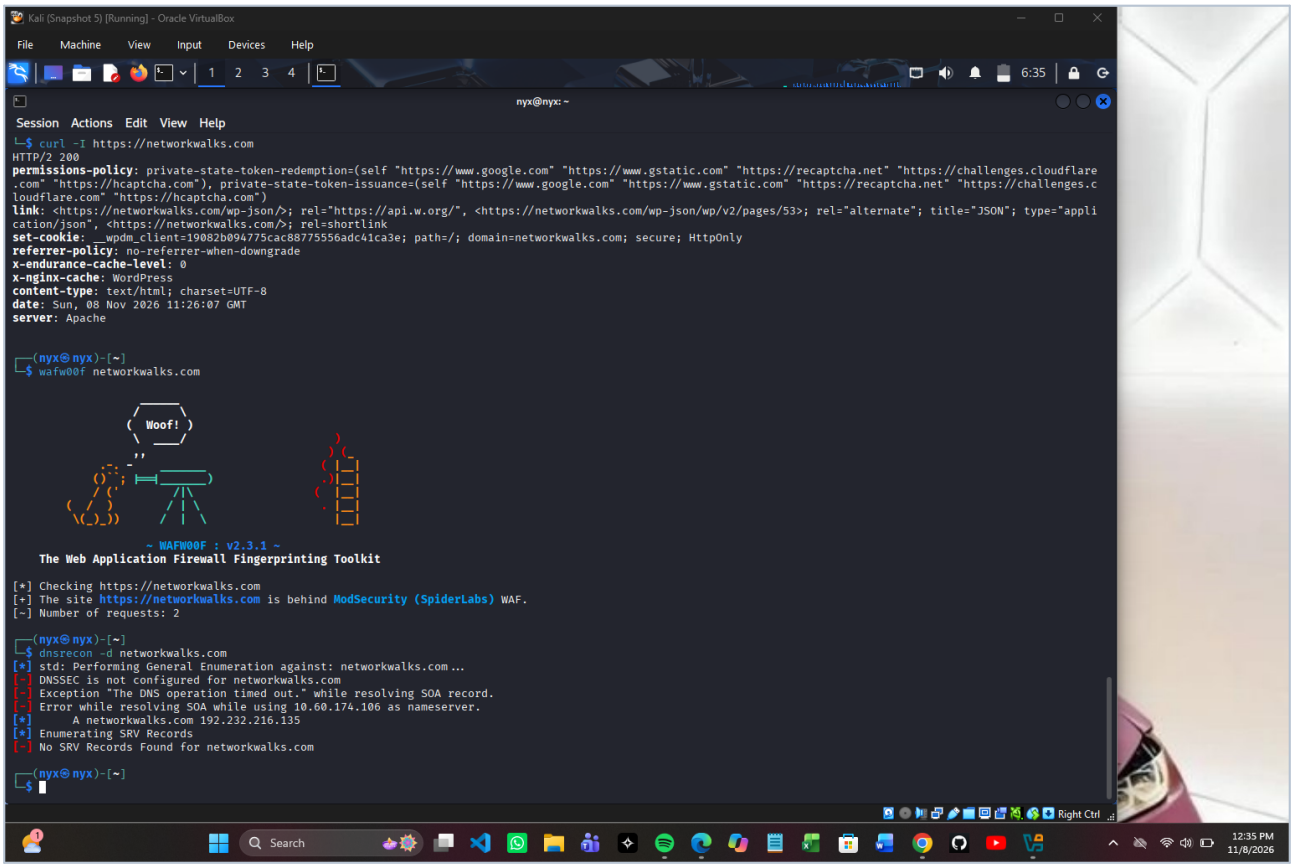


Figure 5: Zenmap ping scan of 10.60.174.0/24 (`nmap -sn 10.60.174.0/24`, 11 November 2026)

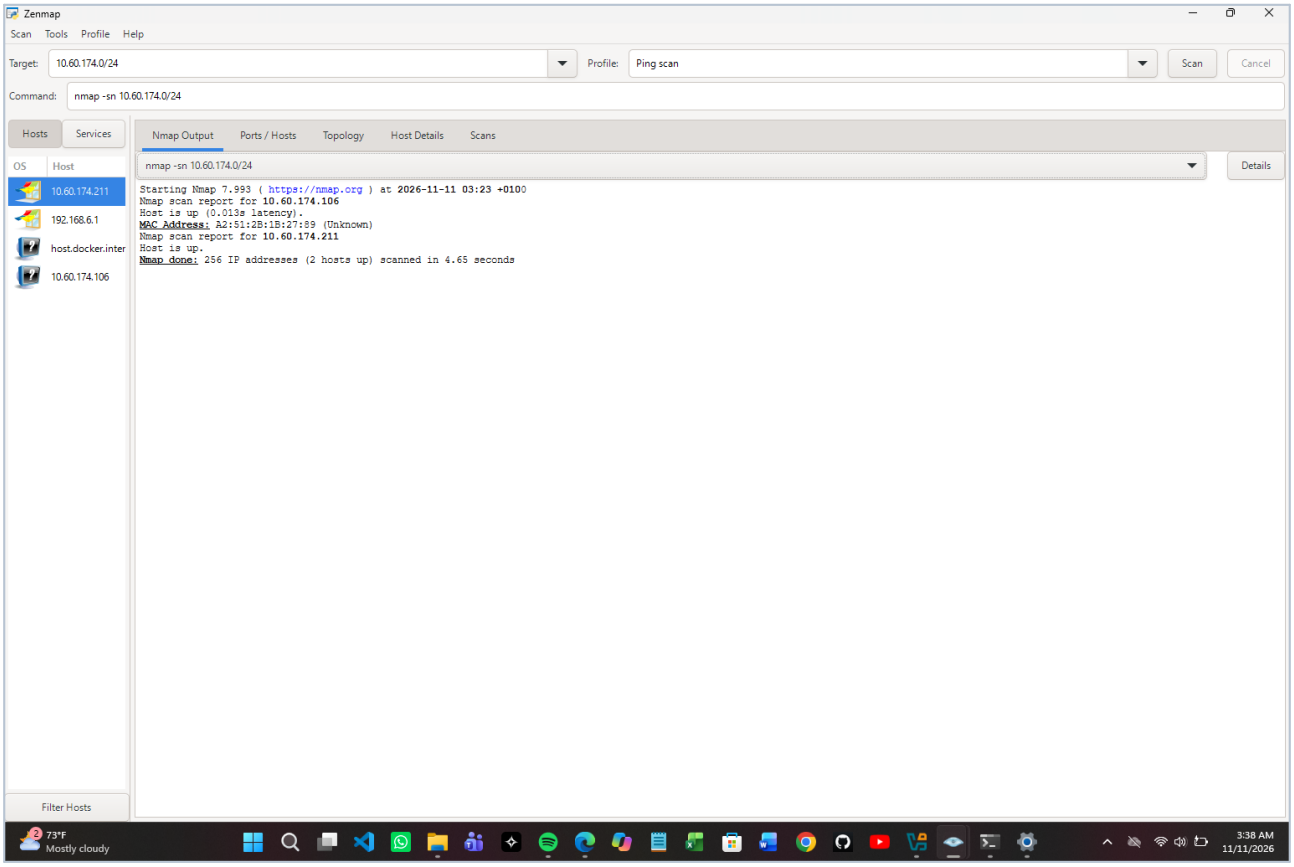
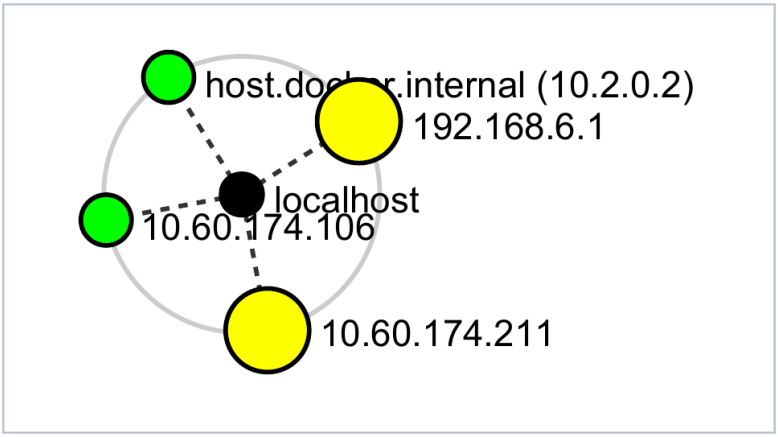


Figure 6: Zenmap network topology (localhost, 10.60.174.106, 10.60.174.211, 192.168.6.1 and host.docker.internal)



-End-

Author

Oluwafunbi Bader
Networkwalks Intern B084
LinkedIn: www.linkedin.com/in/oluwafunbi-bader-167423390

Project Information

Program Name: Cybersecurity Internship Program at Networkwalks | **Week:** 02 | **Repository:** Github